

ActInf Textbook Group ~ Cohort 5 ~ Session 16 (Chapter 7, Part 1) ~ 2/19/24

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 16 (Chapter 7, Part 1) ~ 2/19/2024 | 56:55

all right welcome back cohort 5 we're in
our first discussion for chapter 7 so
Andrew thank you
for facilitating feel free to take it
from
here um so yeah uh first meeting on
chapter seven so I'll do what I did with
chapter six and maybe just give a I'll
attempt to give a brief rundown of the
chapter outline sort of layout and
what's going on um that said there's a
lot going on in this chapter and I would
say aside from chapter six those who are
interested in getting started on
building actor inference models are
going to want to kind of give a a kind
of special amount of weight to this
particular
chapter um we do we are introduced to
Hidden Markov models and partially
observable Markov decision-making
processes which PP is a shorthand for
that and it's it's just as of right now
that's kind of the not completely
ubiquitous but most widely used models
for a variety of Behavioral experiments
um that said U couple weeks from now
we'll get into chapter eight where we
talk more about continuous models and um
that'll be for looking at instances
where we're looking at very you know uh
fast time scales and great for things
like motor movements um and just with
things that are much quicker but usually
we're dealing with discret variables so

um chapter

7 quick intro

7.1 uh we focus on models of categorical variables in discrete time with examples increasing in complexity so we're introduced to models of perceptual processing decision making info seeking learning and finally hierarchical inference um and the authors try to um kind of get at different sorts of emergent properties including measurable physiology and behavior from these models

7.2 were introduced to the hidden Markov model on a perceptual processing experiment um this is where where the the model is of our beliefs about how a musician's audible notes which are the observations or outcomes in the model are generated from a written musical score which are the hidden States the actually written notes um they're uh they simulate the Dynamics of basian belief updating induced by a sequence of her notes again the outcomes

um and then from there it gets into before I move on to the next section it's just worth noting all of these experiments in this chapter more or less to note very specifically what are the hidden States uh what are the outcomes what are our matrices that we're looking at um the a matrix or likelihood Matrix probability of observation given State um our B matrices which is our beliefs about how one state moves to the

next uh from time step to time step we have our D Vector which are our prior beliefs uh about States just starting at time step one that's it once you move on from there uh D is no longer it's in the equation but it's not repeatedly used at every time step um although it can be updated depending on how you're running an experiment such as from trial to trial

um so we get to see how it evolves over time time there are five total time steps it's another thing important to note a lot of experiments you'll typically decide how many time steps occur or will occur over the course of your experiment so here there are five a total of five notes that are played thanks Daniel for going to figure

7.2 um I guess to get a little bit more in depth here uh upper left of that figure shows the the models kind of um posterior beliefs about States and it's confidence and so like these lines start to stretch outward at first it's getting the the right notes for time step one and time step two Suddenly at time step three we get an incorrect note or at least an unexpected note uh there's a difference between the observations which are in the lower right corner um a certain note is played twice in a row incorrectly versus um what should have happened or what we expect expected which is the top right graph unless you can see that lightly gray square that is and the model correctly inferred the right note one that's in black but it

was thrown off by that incor incorrect
note occurring that's the gray there so
the these are all basically outputs of
the the model as the experiment is run
um at the time of this experiment I
believe they were using mat lab and SPM
is their um kind of you know packages
and software used for for deriving this
experiment so there's a lot of
um um nice functions for producing
graphs and outputs of the experiments so
you can actually see what's going on
when we run run them um moving on
to uh section 7.3 now we have what is
arguably a really key sort of model to
dig into this is where decision making
and planning are introduced
into um an experiment and so now our hmm
becomes a
PDP Um this can be seen let's see figure
7.3 um is a great one to it's kind of a
um Exemplar uh image of a of a PDP where
we have many elements as we had before
we have our uh D Vector the initial
States starting all the way to the left
from there we move on we have our states
we have our outcomes s and O we have our
a likelihood matrices which map the two
we have our B matrices showing going
from state to state um but now what
we've done is we've added policies which
are sequences of actions um that the the
agent holds like it could do this series
of actions it could do this series of
actions and so on um those are all kind
of scored so to speak by what's another
quantity that's been introduced here
expected free energy um which differs

from variational free energy that we've seen before. Uh, in that G now involves it takes into account preferences which are also not imaged here but you can see on the right, um, they are denoted in that like list of equations, um, and those impact how G is calculated. That is, you kind of are trying to score. Should I do this to get what I want? Should I do this to get what I want? Should I do this to realize my expected or preferred outcomes or should I do this instead? That is impacting how the agent from there computes its transitions. So what we've done is introduced action into the model where Action Now impacts the environment and it's expected to impact the hidden states of the world that the agent is inferring. So it's just kind of like the prior example but now we've just added this whole new dimension of action where there's an actual bidirectional relationship between the agent and environment. Um, we are also introduced to a couple of concepts whenever the authors break down expected free energy so we have like negative epistemic value and we also have pragmatic value. Um, this relates to the famous exploration exploitation uh debate or dilemma. Um, there's a broad statement in this textbook that uh active inference naturally balances out exploration and exploitation that's its own kind of conversation to have and some people debate that depending on how you want to

Define what balance means the key Point here is that exploration and exploitation for uh EX epistemic value and pragmatic value changing your mind or changing the world um are included within the same cost function if you want to look at this from a machine learning um perspective for example or more statistical perspective or calculus what we're doing is that we're minimizing this quantity that is expected free energy and you're minimize you're minimizing it by engaging in both kinds of behavior both of them contribute to minimizing the same quantity and I believe that is what is meant whenever they state that um active inference naturally balances those two uh kinds of actions that an agent can take we're also introduced to other things that you know for the sake of time I won't go too deeply into but we are introduced to factorization here so there are different kinds of States um that that the agent can try to infer um they're also so different kinds of outcomes or sensory modalities that the agent can try to infer for example um the yes that figure 7.4 uh the A1 Matrix is just mapping uh exteroceptive the exteroceptive modality or set of outcomes which is basically where the rat is in the Maze but then the second Matrix uh a a superscript 2 is denoting uh the rat's beliefs about uh the stimulus if there's no stimulus an aversive stimulus or uh an attractive

stimulus and so the word tensor is used in this chapter and in previous meetings there's been some debate over if this is actually a tensor I believe the idea is that we're ultimately looking at outcomes locations and contexts right so it's this kind of outcomes by locations by context so it does kind of in a way if you attempt to to put it together would be more of a tensor object as opposed to a matrix it's it's a collection of matrices here um but hopefully that helps for anyone who's been wondering uh what that term how how that term's being used um similarly we have multiple B matrices because we also have um multiple contexts uh related to location skipping ahead a little bit we're are at least skimming the rest of this section we have our C vectors which are composed of um preferen preferences about location outcomes um which is C1 uh the rat dis prefers staying in the same spot whenever it starts out uh so it's set to a negative one right um You can you might expect that if you set that to zero and kept them all zero there's a chance the rat might just stay there for the entirety of the experiment probabilistically um negative one kind of gets it moving um C2 meanwhile has to do with the stimulus zero uh it's neutral towards not getting a uh not achieving any kind of stimulus six because it really wants

an attractive stimulus neg -6 it doesn't want the aversive stimulus right it wants the cheese it doesn't want you know a tiny shock or something along those lines and finally Ma just as far as the mathematics go those are all passed through a softmax function um that allows you to both include positive and negative values whenever you're initially defining the vectors um and then also it normalizes everything such that we're looking at more like probabilistic um probability distributions so actually after the math there um for example the attractive stimulus is actually 400 times more probable or we might say more desired than the neutral stimulus um and then again we have our D Vector beliefs about um and again multiple it's factorized there multiple modalities so uh the rat believes that it's starting at the starting location with full probability that's how we said it meanwhile for D2 uh that has to do with where is this where is the stimulus that it wants uh it doesn't know so we have a purely uniform D Vector there uh half and half so um that experiment also includes things like a big part of it is there's a q um in the lower position of the Maze and so this is a great experiment to look at as well because now we get to see both epistemic and pragmatic value epistemic it doesn't know where the reward is uh the Q gives it epistemic value it gives

it information about where the reward is
such as in the image there it's it the
time step one rat starts at beginning
location time step two rat moves to the
Q at the bottom of the maze it uh the Q
tells it that it's at the left position
that's the location of the reward
finally time step three rat smart rat
follows that Q moves to the attractive
stimulus at the left position um which
is pragmatic value realizing its
preference that it does want aches so so
we we we end up with kind of this you
know two-step process where it derives
epistemic value from the que pragmatic
value from actually achieving the the
attractive
stimulus

um so those are some key ideas in active
inference in general and yeah again kind
of an Exemplar model to wrap your head
around um especially for those who are
still kind of trying to understand the
material here because there's a a
significant amount but this is a great
starting point point for really trying
to get into to
modeling um section

7.4 have a model of information seeking
this is a kind of an a cade experiment
um where an agent looks at four
different squares um this does start to
break down things like uh it breaks down
epistemic value further um rather
lengthy explanations of these terms but
but they can be very important and
they're uh very helpful for
understanding the behavior so um those

uh squares there uh all the way to the left this is where all of the squares are given uh equal epistemic value right they're um the agent will look at any of them with roughly the same frequency uh in the middle one however um in that situation the top left square is uh by the person who set up this experiment and its distribution um they gave the top left Square more expected ambiguity which is one of the the terms that's derived from epistemic value um beliefs about that square are less precise they're more uniform uh in the a matrix for this Square the likelihoods uh you can kind of equate this with uh the street light effect the idea that if someone is out at night they drop their keys um they don't know where they are they will probably look where there's more light first so they the idea is oh it's expected that you will actually be able to minimize your uncertainty or reduce your uncertainty by looking at the other three squares here uh whereas the top left one just imagine it's as if that square is is is blacked out or or not well lit or dim or something right there's it's going to be hard to derive any kind of epistemic value from it because it's not things are not very visible um so that's an attempt at getting at how to understand expected ambiguity and finally top right uh or excuse me the rightmost um image uh here all of the squares are still visited um but the B transitions are made to be less

predictable um and so it's uh the lower left yeah the lower left square is given a higher post posterior predictive entropy so the agent is less sure about what what will be seen if the cic is chosen for this square if they keep looking there it's as if it's as if they uh are unsure if there will be new information to get out of that square so they keep returning to it again and again um whereas for the other squares those are more equivalent in the idea that oh you look at them they don't really change why look back again right so it visits all of them but it keeps checking out the lower left

one um we do have box

7.1 which introduces us to it's been mentioned before in the textbook but this is just a quick rundown of trying to describe uncertainty and precision or um priors over Precision so that's a term that basically it it start you it's like game

control uh from the neural process theory that we've looked at uh previously especially in chapter 5 um this is kind of like at amplifying not adding to but rather amplifying certain neural signals um Precision is inverse variance so something has very high variance it means it has low Precision um kind of hard to trust something that has very high variance or very low Precision meanwhile um High Precision means oh that's seems more trustworthy whenever carrying out your computation so it's in a in a sense it's like

waiting uh differently all of these
different objects the different matrices
or particular beliefs or otherwise uh
within the
model um section 7.5 learning and
Novelty um this is where we're
introduced to beliefs about
parameters um so this shows us we're
showing this
on page 141 figure
7.10 or 7.10
um it's basically the same model as
before except now we can see all these
additional
nodes uh where all of our different
objects are
parameterized so our likelihood matrices
the A matrices those are all
parameterized by tiny a or small a um
same thing for the B matrices
parameterized by small B um C Small C
and so on
um those mesh really well with looking
at things like Precision because we're
looking at things like waiting um so
these are all parameters that can be
learned as well over the course of a
series of
experiments
um and typically in active inference
models this uh learning occurs at a
slower time scale um which is also kind
of seen in the final example of the tech
of of this particular chapter um that
has to do with hierarchical inference
but basically these parameters can be
learned as well like how you can learn
over time how much to

trust uh certain likelihood mappings
versus others or
trusting uh your beliefs about
transitions versus others or which um
policies to take versus
others things like that um
brief mentions about other things um
I'll leave
them off here but it has to do with
choosing conjugate priors whenever uh
developing um variables that are linked
together concentration
parameters uh the importance of display
distributions for categorical variables
where observations and states are kind
of counted over the course of an
experiment um the more you see this the
more you expect it to happen it also
helps us to remember that like a model
like how quickly it will update so you
know you flip a coin five times you see
it's always heads uh you might quickly
come to a conclusion that uh it's an
unfair coin meanwhile if you flipped it
a hundred times with a 50/50 of heads or
tails and then you just happen on the
next five tosses to get um all heads
you're much less likely to think oh well
this is an unfair coin because clearly
it was pretty fair the first hundred
times around right so by accumulating
more counts um that informs us about how
much the the model or the agent might
update its beliefs after the next you
know new observation that comes
in we have a mention of structure
learning which is an entire topic on its
own and there's a lot of work still

being

done uh on structure learning but it's rather interesting uh I'll just briefly mention this is seen as something like offline

learning um it there's a metaphor used for it like that occurs during sleep um that is the model or agent is not taking in any new data and yet it's actually updating its beliefs um but more specifically what it's doing is that it's scoring the model it currently has of the world uh let's say and uh during this this sleep or resting offline state it can actually uh compare that model that it has now with let's say a simpler model that maybe has removed some of the certain variables or or or readjusted and minimize some of its matrices or otherwise to see um if it can produce uh an equivalently if not better performing model um that nonetheless has some of its complexity reduced and suddenly it's computationally more efficient right so so basically in a sleeper resting state we have basian model reduction um is is one way that can occur where the model is kind of pruned and and improved um even though there's no uh additional data coming coming in at the moment and then finally um SEC section 7.6 hierarchical or deep inference um this again is very similar to um the PDP that we we saw previously except now we have an entire additional level added above um this is figure 7.12 on page 148 and so um I would say briefly that um the observation nodes uh in the middle

between level one and level two that's a bit of a

misnomer um there what's going on is that the state inference occurring at the lower level is being fed up into level two uh as if they were observations at level two um there I I note this because in a lot of um other papers that I've personally read uh usually they would not include those extra observation nodes in the middle they would just go straight level one up to level two um but this is a nice way of illustrating like it's as if those are observations for level two so what's going on what we have at level one multiple time steps like three time steps uh versus for every single time Step at level two so this demonstrates that there are multiple time scales going on here level one is happening at a faster time scale there three time steps for every individual time Step at level two um the state predictions occurring at level two are being fed into uh the uh computations at level one and so it's like we have these descending um predictions coming from this higher level and then meanwhile level one's State inferences are acting as ascending um observations uh if that's useful and um so this is um I would maybe I'll drop um an article uh link into the chat but I think that there's a rather nice um paper that that I've spent time reading and I I find personal interest in it um but it's it's a paper

called deeply felt affect uh which came out a couple of years ago and it relates to I would say Fields like computational psychology or psychiat um but why I think it's useful for anyone who wants some further reading here is that um it one is a recreation of the tze experiment uh that we saw earlier uh and on the other hand it adds a hierarchical layer um that is theorized in the paper to relate to emotion and so we have a rat that is um given this kind of emotional or affective layer to its model and it actually dramatically changes its um Its Behavior whenever navigating the maze uh it becomes much more confident uh depending on its emotion in its own model of how to achieve the reward um sometimes to the degree that it will skip the epistemic value from the queue it'll go straight to the reward why because now it gets an extra time step that it gets to spend with the reward um so it's it's again it's just we have the teams we have a hierarchical model and then finally we get to see how like Behavior changes whenever we add this hierarchical layer and whenever it comes to cognitive modeling that's a big aspect right it'll be to kind of fiddle around with what variables do you need uh in your model what are the layers to be included um and and how do you kind of tune or prune things from there um I feel like I've spoken more than enough so I will now uh open the floor if

anyone has any thoughts or questions or
just general things you're
considering

um sorry

pach uh actually I have a very basic
question uh since we have been talking
on matrices is the D Vector just like a
onedimensional vector that's it yes
nothing else oh wow so I will say in the
so we saw the te uh experiment for
example and in that case we did have two
vectors

uh one for each uh modality so the first
one was yeah the the rat's belief about
where it starts so the location uh
factor or modality and then D2 its
beliefs about where the stimulus is is
it to the left is it to the right here
it's uh you know hardcoded as uniform uh
as a starting point but yes uh they are
each uh 1D vectors um and that's all you
need to kind of kick off or start the
the simulation whenever you're running
it yes so that is mapped to the number
of factors of the B Matrix
right yes to the edge The Edge length of
B okay

yeah because B maps from hidden state to
Hidden state so whatever the number of
hidden States is that b is just that per
slice and then the number of slices of B
is equivalent to how many affordances
there are and then D is just the edge
length that's carried through with s and
then a is $O * s$ because it Maps o and
s pi is also a list with the
probabilities of the actions in the
policy space summing to

one thank you

yeah

I have one question

sorry uh in like actually like when you normally try to initialize these vectors say uh say d can you just also hardcode it like just like one and then just pass it

right

yeah yeah that's right you would you would hardcode it um that's one of the kind of theoretical discussions around active inference right is it's it's more about the what and the the why and the how but not so much about how it starts um that said what you can also do as far as learning and inference goes is you could have um you could set up an experiment that runs over multiple trials uh in that case you would still hardcode D but going from trial to trial you could have it such that the agent actually updates uh deep over time so that trial one it begins with those hardcoded values trial two it might update um you know now that it's the start of trial two what is what are its prior or initial beliefs about the state yeah yeah so like in this case let's just say it was GNA do three rounds of the

teamas and then initially it has a a flat uh you know it has a certain it believes it starts in a certain setting that that could be that that would be the D1 location or maybe that one stays the same the one z00 Z and then the one it's like it learns by the end that it

it it always starts on the right or
updates it towards the fact that it's
that the reward of the other the stimul
food is on the right side it could learn
that by just counting up okay it was on
the right it was on the right it was on
the right and than that kind of like
putting Pebbles of different color into
the N that's the learning by
counting which is reflected here with a
little D over
the learning by counting on this one so
this one starts out Z 0 it starts out
one one and then you add one every time
you observed it and then that ends up
that's a hyperparameter on the D
distribution so hence the kind of
strategy of the book of showing like the
minimal figure 4.3 type model and then
here's like one Motif which is hyper
priors for learning or attention like
Andrew mentioned here's another Motif
nested models but the hyper priors are
not
shown you can start to combine these but
then you can also imagine how the models
complexity grows a
lot sorry what do you mean by motive
like here The Motif that it's showing is
just nesting it's not including Little D
so it's not like it this is not a
generalization of the previous
one this is just like a generalization
of the of the figure 4.3
one
as this one is but this one's not
nested so like here but the difference
that it's trying to show is just about

the idea of putting this the metallic
lowercase above each of
letters a b c d
e and then these motifs get kind of
combined
okay
um question
yeah slightly slightly tangential so
Andrew in the react FX um project that
you're doing are you building these
types of models I know it's very simple
question um well we've just kicked off
the the project a couple weeks ago so
we're still kind of trying to
decide what kinds of models to build but
anyone who's interested can can
certainly you know hop on and and see
what's going on we do have a a distinct
Coda for that and so right now people
are running project ideas and so they're
like in a table that you can just put in
like oh I would like to try doing
something like this and then maybe put a
couple notes on on what direction you
think it might take and um yeah as
Daniel's pulled up here uh for me I
wanted to see if we could do kind of
reproducibility project where we take
some of this you know few years old code
in mat lab or SPM can we run it through
uh Julia and RX and fur instead um just
basically to see if we can to help with
reprodu reproducibility and from there
we could even potentially start moving
into like modifying those models or
maybe improving
them yeah generally I would it seems
like pdps might be one of the primary

ways that we go so hopefully that
answers your question and is that all I
I see Daniel posted the links in the
chat and I I'll take a look um but is
that all happening at this point
asynchronously through Koda or are there
um Thursday meetings Okay this what you
got up here on the and we're developing
out the GitHub repo and so on so
there'll be but there's there's more
than ample sync and synchronous
opportunities if people want okay and
and it's I mean and there's no chapter
six for RX and fur but these are some of
the um it um I mean we can go more into
this another time but we've like talked
about how we can best contribute and
right documentation so for people right
so it's more I mean it's still sorry
sorry but it's it's still the underlying
strategy yes
we're learning this package for
generative model that's also over the
course of the Year going to increase a
lot in its capacity and
usability okay cool all right I'll pull
up that Cota thanks a
lot and just to note it'll also be
incorporating um you know some research
that's been done since the publication
of the textbook we're looking at so um
you know there'll be a little bit more
uh up to dat so to speak the the
textbook is still a Cornerstone of of
active inference but that said there
than some you know certain kinds of
updates that have been made since then
yeah I would just I would just love to

see this um those modeling applied with
you know actual values if you will it
will help me wrap my mind around it cool
absolutely um I did notice uh Susan
posted a question in the chat how would
reflection versus indecision be
represented um

I find that to be a rather interesting
question myself and uh while that's not
directly answered by the textbook uh
surely there are papers that have been
published on the these kinds of topics
um but in any case intuitively I would
say just to kind of stay with the
chapter

itself uh

chapter excuse me uh page 145

146 um this is uh an agent navigating a
maze and in this case

um uh the agent is does not have any
specified like particular preferences or
rather that it's like purely uniform
preferences and so the a it's as if the
agent doesn't prefer anything in
particular uh whenever the model is set
up in that way then the agent uh engages
in what could be viewed as an emerging
Behavior called novelty seeking where it
just keeps going to places that it
hasn't been to yet uh you know there
there's no preference for a certain
reward or stimulus to where it knows
that it wants to go to a space that has
a reward right so it doesn't go there it
just goes everywhere um that said it's
still doing it's still following the FP
it's still minimizing expected free
energy or variational free energy and so

uh it will avoid spaces it's already
been to because it's already been there
um nothing new to get no new epistemic
value to be derived from a space you've
already been to and so it will just kind
of like run the full gamut of all the
spaces it's never been to uh this is a
phenomena that's also been referred to
by pner in 1985 is the inhibition of
return so that's that's a pretty
interesting thing to see and it makes a
lot of sense because when we look at our
equations C uh C vector or preferences
are uh heavily informing G and
so there's nothing informative about SE
because it's purely uniform you know uh
nothing nothing's particularly preferred
then it doesn't really help with you
know figuring out what to do other than
just keep keep running around getting
getting all that novelty you can U maybe
it's thrill seeking um it's a joke but
reflection meanwhile not so sure
intuitively might have a relationship
with structure learning um you know an
agent doesn't necessarily have to be
sleeping in order to uh kind of tune or
prune its model um but the idea is that
yeah there's no incoming sensory data so
reflection might relate to that process
you know someone who's sitting there
reflecting uh kind of thinking through
things through even though they've not
pres been presented with anything new um
might be a for struct learning there
might be a paper out there um or or a
dozen uh on these kinds of things worth
worth uh searching for that um but yeah

very interesting question

um yes uh

Anna I see put you're muted

MD

can you hear me okay now yes okay okay

sorry about that uh yeah I was wondering

if you could elaborate on a little bit

on

um the offline updating of of

beliefs um and I'm curious about the

scientific studies that have been done

demonstrating that this in fact happens

that you can you're you know during

sleep or maybe meditation or daydreaming

right can you elaborate a little bit on

how active inference scientists have

looked at this

aspect about active

inference so just be direct in saying

that yeah I'm not personally too sure of

the research that's been done very

specifically on structure learning in

related in relation to

actual sleep um or meditation I mean I'm

seeing just from a a quick all show one

this is not exactly AC but it's it's B

it's right there I'll put it in the

notes this is kind of a g there's many

similar architectures done in machine

learning where there's this

phasic um

approach W with extending and updating

during the day with a fast inference and

then um using different techniques for

sleep and I mean it it probably gets

into the specifics of the machine

learning architecture and then also

there's the whole other angle you said

like

empirically um what what's happening at
the synoptic level and at like the
cognitive level for humans and skill
consolidation and sleeping learning
consolidation naps there's there's
probably a lot of different phenomena
that thank you

just drop this here but this looks to be
a manuscript by a couple of researchers
as well who uh make a mention of
particular meditative

Traditions um but they claim there that
as opposed to it being um structure
learning per se they they claim that
meditation is more of a form of like
it's it's a particular It's A peculiar
policy itself um that is you're choosing
to do that um that actually looks to be
a really interesting paper and I
appreciate that there's a more U kind of
culturally accurate as opposed to highly
abstract um kind of theme to it so um
Susan so yeah this is fascinating um
thank you Andrew

so what uh what keeps coming up is is is
the reference to
you know learning is takes time or um or
you there's no fast learning and so
I'm I'm wondering if maybe the concept
of learning

itself is you know is is um more
specific than I guess what we um what we
generally think about learning so I mean
how would you
categorize learning or or well maybe why
that why it keeps referencing that
learning is

slow slower than

what yes this is a great question I'll
give a first thought um especially as as
more complex cognitive models are built
there's several ways that different time
scales of adaptation happen so the
fastest layer is called inference
and that's kind of like the rubber hits
the road like each um new observation
coming in but also the whole process is
called inference um but also the fastest
layer is called inference then um the
next slowest layer is attention so like
attention modulates um on over multiple
sensory Cycles like just thinking about
the visual case or the the dinner party
like overhearing multiple conversations
like attention module

faster than um slower than the words
coming in but um faster than updating
like deeper factual beliefs and then
then

learning um as distinguished from those
other kinds of parameter
changings is like here being modeled
explicitly with these Hypo priors and
updating the distribution that the kind
of rubber hit the road parameters are
being drawn from that's not the only way
to model

learning so hence chapter six and really
digging in with like what time scale
analysis is being studied because maybe
whether it's learning over seconds like
uh repeating back a phone number or
whether it's like a um semester level
maybe that's the time scale of your
model so then that learning is happening

on the kind of primary time scale of the model

it's just way too General to say without specific system of interest but that being said there are multiple ways that like to kind of and the hierarchical models like looked at so multiple non-exhaustive or exclusive ways to model the multiscale temporal dynamics of

change yeah I I would definitely second uh Daniel's point about yeah it's it there is no single answer to the to the question again we have this final example in this particular chapter referring to yeah words and sentences right so those time scales don't seem dramatically different actually right like we're looking at a matter of let's say milliseconds versus seconds um like predicting what the next few words are going to be an entire sentence to try and capture something like what the sentence is about versus the the faster time scale of word word word word like word to word um in the in the sentence but I I did want to mention a couple other uh as far as like um trying to map this to uh Concepts or behaviors in Neuroscience um another way of thinking not necessarily thinking about it this way but one potential analogy could be something like working memory versus versus long-term memory uh right like like here's your quick access like working memory memory of the Moment by moment long-term memory occurs over time you know it gets

Consolidated um slowly over time and it
it can lead to long-term changes in
things like uh your daily habits um what
comes more naturally or quicker to you
so it just describing it a bit more
intuitively that way yeah I like to
sometimes think of it as like yeah
shortterm versus long-term memory then
in addition to that as we saw in chapter
five uh in terms of the neurobiology
um it's commonly been found or or
theorized from from various studies when
working with neuroimaging data that uh
whenever a participant in these studies
is learning something uh what happens is
that that you tend to find when looking
at brain rhythms from let's say EEG um
that uh more Theta and gamma waves are
present simultaneously and so they view
it as what would be called a Theta gamma
coupling we do get a reference to that
in chapter 5 in this textbook um Theta
rhythms are slower uh in a in a
particular kind of Rhythm band or
frequency band uh gamma are faster and
so what happens is these rhythms kind of
um you know it's kind of fun you can
think of it as like music you know you
have some fast drums and a slow piano or
you know some such thing but there's a
kind of sinking between these two uh
differing rhythms and it it is seen as
refle um being a kind of neural
correlate to someone learning something
um so multiple time scales in a PDP
multiple time scales in how our memory
Works multiple time scales in Neuro
Imaging data just a couple more uh kind

of metaphors is Food For

Thought So one one more question that

um it almost I think I'm wrong but it

almost makes me want to wonder um if um

if the learning in this sense isn't

um is it an effect

um

[Music]

um actually acting on a

policy but that may or may not be

correct but the other thing that comes

to mind

um would be

would would

surprise occur at either level in other

words if they if they don't learn or if

their policy doesn't um get the expected

payoff then you know there would be a

surprise but also learning would could

actually Encompass a surprise as well

so makes me want to say that there's a

almost a surprise

tolerance yeah like thinking back to the

chapter 2 the basian surprise is the

Delta of

learning and so if if the hyper priors

are

fixed the basing surprise is going to be

zero no matter how surprising in the

first sense the observation is like we

think that the room is this temperature

and we're not open to learning we' have

fixed hyper prior and then even shocking

observations just continue to shock US

versus if you had um like a kind of

adaptive learning rate then there would

be some like a surprising observation

would come in that would catch your

prior wherever it initially was and then that would also move your prior some amount over to the new observation so if that kept happening eventually like you would converge so that that was no longer surprising and then how fast you converge is attention slash the learning capacity over different time scales so learning capacity would come into the level of Tolerance I guess would coincide thank you and uh forgive me um to clarify so learning capacity is represented through those hyper priors yeah um indirectly yes like like when Andrew mentioned the coin flip it's like when um and this is just um you could see this in any kind of uh Bayesian statistics overview like if you only had flipped the coin twice and you got one in one then you'd have a wide prior centered around 50% because your maximum likelihood estimate for the fairness of the coin is 50/50 but it's very dispersed it can't be zero can't be 100 because you know that you've seen at least the other things but but it's broadly like it's it's a very over dispersed and then if it was 1 million and 1 million then the the the two millionth and one flip would be very tight would only move the distribution a tiny amount so that's like that learning by counting and then as you count up more and then so then then you have this whole question of okay well then but

then that just slows down our learning rate but maybe it's a cell differentiating and it does slow down its learning rate to a halt or maybe there's another process that takes a look at that um learning by counting variable says when this is 100 we cut it down to 20 and and that's this whole challenge with the iterated modeling is figuring out all these like second order rules and diagnostic to be like okay should we cut out 20 or like 30 and then those are all the like simulations and sweeps that you do and then investigate how the agent is behaving in different regularities got you thank that's a good explanation that's a lot of compute it is so even that's one very interesting thing which is like even when like the model as parameterized might be low compute to run the open question is how much did it take to get there not dissimilar to like a language model where in the forward running of a trained model it's not Mega computationally expensive but then training it can be expensive and similarly while updating the model might not be expensive so for a toy model we're just whatever you make you let it run and it's all good but to find like adaptive and resilient ranges of parameters with many many possibilities that can be running these models many

times cool

well we

return for chapter 7 part two next time

or feel free to

join chapter two it's kind of funny

chapter 2 and chapter 7 they

are very similar and in Chapter 2 we

kind of reference chapter 7 because

that's where the discreet time models

are then chapter 7 took us back to

chapter 2 with the surprise and

everything

so oh thank you Cheryl I'm going to copy

this

in you're welcome thank

you okay farewell bye he bye bye every